Multimodal Machine Learning for Explainable Disease Risk Prediction and Clinical Decision Support

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Machine learning enables disease-risk prediction by learning relationships between demographic, clinical, lifestyle, and longitudinal healthcare data to identify individuals at elevated risk and support early intervention. Effective predictive systems generally involve data preprocessing, missing-value handling, feature selection, and classification using algorithms such as Random Forest, Logistic Regression, SVM, XGBoost, and deep neural networks. Deep learning and state-space approaches can further model complex and longitudinal patient information contained in electronic health records and medical notes. Explainable AI techniques such as SHAP and LIME improve transparency by identifying the contribution of individual features to predictions, supporting clinician trust and decision-making. However, practical deployment remains constrained by data quality, label noise, overfitting, limited generalizability, privacy, and insufficient clinical validation.

I. INTRODUCTION

Early identification of disease risk is an important component of modern healthcare, as timely prediction can support preventive measures, patient stratification, and clinical decision-making. The increasing availability of clinical records, demographic information, lifestyle attributes, laboratory measurements, and other real-world healthcare data has created opportunities for machine-learning techniques to identify complex patterns associated with disease development. Traditional approaches such as Logistic Regression, Support Vector Machines, and Decision Trees provide efficient baseline models, while ensemble methods including Random Forest, XGBoost, and advanced deep-learning techniques can capture nonlinear relationships among heterogeneous health-related features. However, the effectiveness of disease prediction systems is influenced by challenges such as limited datasets, noisy or incorrectly labelled medical data, feature redundancy, class imbalance, overfitting, and difficulties in generalizing models across diverse patient populations. Furthermore, the black- box nature of complex predictive models can reduce confidence in their use within healthcare environments, creating a need for transparent and interpretable predictions. Recent research also demonstrates the potential of multimodal and longitudinal healthcare modelling for representing patient conditions more comprehensively through clinical records, medical notes, and lifestyle information. Despite these developments, practical clinical deployment remains challenging because of data privacy, model robustness, generalizability, transparency, and the requirement for clinical validation. Therefore, there is a need for an intelligent, robust, and explainable disease-risk prediction framework that can integrate relevant healthcare information and provide reliable predictions while supporting healthcare professionals in making informed decisions.

The main contributions of this work are:-

1) A machine-learning-based disease risk prediction

2) framework integrating relevant clinical, demographic, and lifestyle features for early identification of individuals at elevated disease risk.

3) A comparative predictive modelling approach evaluating multiple machine-learning and deep-learning algorithms, including Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and neural-network-based models, to identify effective approaches for disease-risk classification.

4) An explainable AI layer using SHAP and LIME to provide feature-level explanations for individual predictions, improving model transparency and supporting healthcare decision-making.

5) A robust prediction strategy for healthcare data challenges, considering issues such as label noise, data quality, overfitting, class imbalance, and model generalization across different patient populations.

6) An evaluation framework for real-world applicability, emphasizing predictive performance, interpretability, robustness, privacy, generalizability, and the need for clinical validation.

II. RELATED WORK

Machine-learning-based disease prediction has been extensively investigated using clinical, demographic, lifestyle, and real-world healthcare data. Traditional approaches such as Logistic Regression, Support Vector Machines, and Decision Trees provide computationally efficient baseline models, but their ability to capture complex nonlinear relationships can be limited. Ensemble approaches including Random Forest, Gradient Boosting, XGBoost, and related methods generally provide stronger predictive performance by combining multiple learners and modelling heterogeneous healthcare features. Recent research has also explored deep-learning approaches for disease-risk assessment. Niu et al. address the difficulty of modelling longitudinal electronic health records using a deep state-space model that integrates patient latent states, medical notes, and predictive clustering for personalized disease-risk prediction. Other studies employ feature-selection and optimization techniques before deep-learning prediction, showing that selecting relevant attributes can improve predictive performance while reducing computational complexity. Multimodal approaches further combine clinical and lifestyle information because disease risk is influenced by multiple physiological and behavioural factors; however, feature alignment and preprocessing remain important challenges when heterogeneous datasets are integrated.

Explainable AI represents another important direction in disease prediction. SHAP and LIME have been applied to identify how individual input features influence automated predictions, improving transparency, trust, and interpretability for healthcare professionals. Recent multimodal explainable systems combine algorithms such as Logistic Regression, Random Forest, Gradient Boosting, and XGBoost with SHAP and LIME and provide web-based interfaces for obtaining interpretable risk predictions. Systematic evidence from real-world healthcare studies shows that Random Forest, Logistic Regression, and Support Vector Machines remain among the most frequently used machine-learning techniques, with electronic health records, patient registries, and wearable devices serving as important sources of real-world evidence. However, many existing systems continue to face limitations involving single-dataset dependence, generalization across populations, interpretability, privacy, and practical clinical deployment.

Therefore, existing research demonstrates that combining robust predictive modelling, heterogeneous healthcare features, explainability, and practical deployment can address several limitations of isolated disease-prediction approaches.

III. PROBLEM STATEMENT AND OBJECTIVES

Individuals and healthcare professionals require reliable tools for identifying disease risk at an early stage, but existing prediction approaches face several concrete challenges: (1) dependence on limited or single datasets that may restrict generalization across diverse populations; (2) inconsistent healthcare data containing missing values, irrelevant features, and noisy or incorrectly labelled observations; (3) difficulty in interpreting complex machine-learning and deep-learning predictions; (4) limited integration of clinical and lifestyle information despite their combined importance in disease-risk assessment; and (5) insufficient translation of research models into practical, user-friendly decision-support applications.

Existing predictive models address individual aspects of these challenges, but a unified framework combining prediction, robustness, explainability, and practical decision support remains necessary. The project objectives are to:

• Build a preprocessing pipeline for cleaning, transforming, and preparing heterogeneous healthcare data for predictive modelling.

• Identify relevant clinical and lifestyle features through appropriate feature -selection techniques.

• Develop and compare multiple machine-learning models for disease-risk classification and select an effective predictive model.

• Integrate explainable AI techniques to provide understandable feature-level explanations for individual predictions.

• Design a practical prediction interface through which users or healthcare professionals can provide patient information and obtain risk predictions.

• Evaluate the proposed system using appropriate classification metrics while considering robustness, generalization, interpretability, and real-world applicability. The primary target users are healthcare professionals and clinical decision-support users, with patients representing the intended beneficiaries of earlier risk identification. Design-stage success criteria include reliable predictive performance, interpretable model outputs, consistent preprocessing, comparison across multiple algorithms, and an accessible prediction interface. Clinical validation and broader real-world deployment remain future requirements, as existing literature identifies these as significant barriers to practical healthcare adoption.

IV. PROPOSED SYSTEM ARCHITECTURE

TABLE I
TECHNOLOGY STACK

Layer	Technology	Role
Backend	Python 3.11, FastAPI	Async REST API, recommendation and CV pipelines
Frontend	TypeScript, React/Next.js	SSR dashboard with independently refreshable pa-els
Async tasks	Celery + Redis 7.0	CV verification job processing
Database	PostgreSQL 15 + PostGIS 3.3	Core store, spatial (GiST) queries
Storage	S3-compatible object store	Growth-update image storage
CV libraries	OpenCV, PyTorch, YOLO	Manipulation, detection, classification
Auth / infra	JWT (RS256), Nginx, Docker	Stateless auth, SSL/routing, orchestration

The proposed disease-risk prediction system follows a layered architecture comprising the user interface, data-processing layer, feature-engineering layer, machine-learning layer, explainability layer, prediction service, and result-presentation layer. The user interface collects the required clinical, demographic, and lifestyle attributes and transfers them to the backend prediction service. The preprocessing layer performs data validation, missing-value handling, encoding, normalization or standardization where required, and preparation of features for model inference.

The feature-engineering layer selects relevant attributes and constructs the final feature representation used by the trained predictive models. The machine-learning layer contains multiple candidate classifiers, including Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and neural-network-based approaches, enabling comparative evaluation and model selection. Ensemble methods are particularly suitable for heterogeneous healthcare information because they can model nonlinear interactions between predictors.

The explainability layer is coupled with the selected prediction model and uses SHAP and/or LIME to determine the contribution of individual features to the generated risk score. The prediction service returns the estimated disease-risk category together with the corresponding explanatory factors rather than presenting only a classification result. The architecture can additionally maintain model-performance records, preprocessing configurations, and prediction logs for evaluation and monitoring. Security and privacy considerations apply across the system because healthcare datasets contain sensitive patient information; research has specifically identified privacy, confidentiality, and secure distributed learning as important requirements for large-scale healthcare prediction systems. The overall architecture therefore integrates data preparation, feature selection, predictive modelling, explainability, and user-facing decision support into a unified disease-risk prediction workflow.

V. METHODOLOGY AND ALGORITHMS

Environmental parameters - including Air Quality Index (AQI), PM 2.5, temperature, humidity, soil pH, soil type, sunlight hours, and cumulative rainfall - are dynamically retrieved via OpenWeatherMap [10] and World Air Quality Index APIs [6] per queried geographic location and cached per spatial tile [12]. Pollution-severity classification and climate-zone categorization are computed once during the ingestion phase. Because station-level pollutant readings vary sharply with local meteorology, as demonstrated in machine-learning studies of urban air-monitoring networks [15], the spatial cache is tied to a localized tile grid rather than a generalized city-wide average. Candidate plant species incorporate species-level (rather than generic genus-level) attributes, such as temperature and soil-pH tolerance, space- type compatibility, pet toxicity, seasonal allergen modifier, pollution-absorption rate, maintenance level, and growth velocity, sourced from curated ecological reference datasets [7].

A. Recommendation Engine

A strict hard-constraint filter first eliminates any species that fails temperature range, soil pH, spatial capacity, toxicity, or allergen thresholds. Excluded species are permanently removed from the candidate pool, mirroring screen-before-score multi-criteria evaluation pipelines [13], [14]. Surviving candidates are scored using the weighted function:

$$\text{Score} = 0.35P\text{abs} + 0.25(1 - M\text{req}) + 0.20G\text{match} + 0.20E\text{fit}$$

where abs is the normalized pollution-absorption rate, req is the normalized maintenance requirement, match represents growth-rate suitability, and fit denotes overall ecological fit. All metrics undergo min-max normalization [13]. If all candidate species fail the hard-constraint phase, the engine returns an empty result set alongside explicit failure reason codes rather than relaxing boundary parameters.

B. Growth-Update Verification

When a user submits a plant growth image, the Haversine distance between the image's embedded GPS coordinates and the registered anchor position is calculated. Submissions exceeding distance tolerances are immediately rejected. If EXIF GPS data is present and deviates from the reported GPS coordinates by more than 200 m, the request is routed to manual administrative review to flag location-spoofing anomalies [18]. Passing submissions enter an asynchronous processing queue (Celery + Redis 7.0) handled by Computer Vision (CV) workers. These workers execute Error Level Analysis (ELA), plant presence detection [8], growth-stage comparison against prior verified submissions, and species continuity checking, drawing on CNN-based plant analysis paradigms [4], [9], [11], [23] and metadata-coupled classification frameworks [22]. A growth update is marked fully verified only when both spatial alignment and computer vision validation steps pass.

C. Reward Eligibility and Ledger

Upon successful completion of the verification workflow, the reward service requests a row-level database lock on the specific (user,plant) pair and evaluates the elapsed time since the previous transaction against a trigger-specific minimum time interval. If this temporal constraint is violated, the transaction aborts. Otherwise, earned points are computed against the prior snapshot balance, and an append-only row is committed to PostgreSQL 15 before publishing a reward-credited event, ensuring no user is notified of an unrecorded balance update.

VI. NON-FUNCTIONAL REQUIREMENTS AND FEASIBILITY

Performance Targets: 3-second recommendation/dashboard response time, 2-second acknowledgment for growth submissions (excluding asynchronous CV execution), 5-minute CV completion SLA at the 95th percentile, 1-second spatial query response using PostGIS 3.3 GiST indexing [24], and throughput capacity for 200 concurrent

requests/second. Security Specifications: Stateless JWT authentication (RS256) with 15-minute access and 7-day rotating refresh tokens, TLS 1.2+ in transit, AES-256 for data at rest, bcrypt/argon2id password hashing, and server-side EXIF metadata stripping (preserving only validated GPS fields). Reliability Benchmarks: 99.5% uptime target, exponential backoff retries (up to three attempts) with manual fallback for CV tasks, write-before-publish event ordering, and daily database backups with a 30-day point-in-time recovery window. The system utilizes an open-source technology stack: Python 3.11/FastAPI for async backend operations, React/Next.js for the frontend dashboard, S3-compatible object storage for image files, and Docker/Nginx for orchestration. For 1,000–5,000 active users, estimated infrastructure costs range from USD 20–50/month, with a modular layout that localizes worker failure without impacting core API availability.

VII. RESULTS, COMPARATIVE ANALYSIS, AND LIMITATIONS

TABLE II
COMPARISON WITH RELATED SYSTEMS

System	Data Used	Method / Model	Key Strength	Limitation
Steinfeldt et al. [1]	Medical history from EHRs	Deep neural network for phenome-wide prediction	Predicts a large number of diseases simultaneously	Requires large-scale EHR data; computationally intensive
Nagaraju et al. [2]	Clinical attributes	Deep Red Fox Belief Prediction System (DRFBPS)	Optimized feature selection; high prediction accuracy	Focused only on heart disease; limited dataset scope
Niu et al. [3]	Longitudinal EHRs and medical notes	Deep state-space model with predictive clustering	Captures temporal patterns; learns latent patient states	Complex model; higher training and inference time
Alhumaidi et al. [4]	Real-world data from multiple sources	Various ML models (Random Forest, LR, SVM, etc.)	Comprehensive review of ML methods and real-world data	Heterogeneous data quality; limited model explainability
Alzboon et al. [5]	Pima Indians Diabetes dataset	ML models (NN, RF, DT, SVM, NB, etc.)	High accuracy using Neural Network model	Dataset is relatively small; less generalizable
Lin et al. [6]	Stroke screening dataset	Robust learning with dynamic truncated loss model	Robust to label noise; improves learning stability	Requires proper noise estimation; more complex
Halder et al. (ML-CKDP) [7]	CKD clinical dataset	Ensemble models (RF, AdaB, XgB, GB, etc.)	High accuracy; includes web application	Specific to CKD; not multimodal
Proposed System	Clinical + Lifestyle + Demographic (multi-source)	Hybrid ML framework with XGBoost + Explainable AI (SHAP, LIME)	Multimodal data integration; high accuracy; explainable predictions	Needs larger diverse datasets and clinical validation

This architecture is currently specified at the design and system-definition stage; no live field deployment, populated dataset, or model training run has been executed. What can be reported is structural integrity: the two-stage recommendation pipeline strictly prevents disqualified candidates from surfacing regardless of score, and the dual-layer verification protocol mandates independent spatial and computer vision confirmation before emitting reward tokens. Coupled with row-level database locking around eligibility checks, double-crediting for a given time window is architecturally precluded - a property to be further evaluated under high-concurrency stress testing.

VII. LIMITATIONS

Stated limitations of the proposed disease-risk prediction system are: no clinical field deployment or prospective user evaluation has yet been conducted; model performance may be affected by dataset size, class imbalance, missing values, and noisy or incorrectly labelled medical records; and models trained on individual datasets may have limited generalizability across different patient populations. Existing studies also identify challenges related to feature alignment when combining heterogeneous clinical and lifestyle datasets, computational requirements of complex deep-learning approaches, and the need for standardized evaluation protocols. Explainability techniques such as SHAP and LIME improve transparency but do not by themselves establish clinical validity or guarantee reliable predictions across all populations. Privacy and confidentiality remain important considerations when processing sensitive healthcare information, particularly for systems intended for large-scale or multi-institutional deployment. Therefore, reported accuracy or other evaluation metrics should be interpreted as model-development results rather than evidence of clinical effectiveness unless validated prospectively in real healthcare environments.

VIII. CONCLUSION AND FUTURE SCOPE

The proposed disease-risk prediction framework addresses important limitations identified in existing healthcare machine-learning research by combining clinical and lifestyle information, predictive modelling, feature selection, and explainable AI within a practical decision-support architecture. Existing studies demonstrate that heterogeneous healthcare information can provide a more comprehensive representation of patient risk, while ensemble and advanced learning methods can model complex relationships among multiple risk factors. The incorporation of SHAP and LIME further enables the system to provide interpretable explanations rather than relying solely on black-box risk predictions. The proposed system is intended to assist healthcare professionals in identifying high-risk individuals and supporting informed preventive decisions rather than replacing clinical expertise. However, the system remains subject to the limitations of its training data, model generalization, robustness, privacy, and clinical validation requirements. Future work centers on expanding the training datasets with diverse patient populations; collecting prospective clinical data; improving robustness against label noise and class imbalance; evaluating additional deep-learning and ensemble architectures; developing stronger feature-selection and multimodal learning strategies; and conducting clinical validation through real-world deployment. Longitudinal EHR modelling and multimodal healthcare information can also be incorporated to represent changing patient health states more effectively.

Further extensions may include federated learning for privacy-preserving multi-institutional training, uncertainty estimation for communicating prediction reliability, integration with wearable and remote-monitoring data, and improved explainability mechanisms for clinical decision support.

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